

Whole-Number Division: Quotients and Remainders

Answer Key

Grades 4–5

Quick Answers

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|------------|------------|----------------------|
| 1. 6, 3 | 5. 6, 5, 7 | 9. 1300, 260, 269, 2 |
| 2. 7, 3 | 6. 8, 2 | 10. 13, 5 |
| 3. 7 | 7. 24, 1 | |
| 4. 600, 75 | 8. 21, 3 | |
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Curriculum

Level: Grades 4–5

4.NBT.B.6 – *problems 1, 2, 3, 4, 7, 8, 9*

4.OA.A.2 – *problems 5, 6, 10*

Difficulty 1–5 of 5 across 23 tagged checks.

Common wrong answers

8. *If they got 20: stopped one group early, leaving a remainder bigger than the divisor*

1. Circle equal groups: 27 counters, groups of 4. Count out four counters at a time and ring them: after six rings $6 \times 4 = 24$ counters are inside rings. That leaves $27 - 24 = 3$ counters outside, and 3 is smaller than 4, so no seventh ring can be made. Check by multiplying back: $6 \times 4 + 3 = 27$. Quotient 6, remainder 3.

2. Build an array: 38 tiles in rows of 5. Each complete row uses five tiles. Seven rows use $7 \times 5 = 35$ tiles, which leaves $38 - 35 = 3$ tiles — not enough for an eighth row, because a row needs five. The array is therefore seven full rows with a short row of three underneath. Check: $7 \times 5 + 3 = 38$. Quotient 7, remainder 3.

3. Check with multiplication: $6 \times \square + 3 = 45$. The three leftover rolls never went into a bag, so first take them off the tray: $45 - 3 = 42$ rolls were bagged. Each bag holds six, so the number of bags is $42 \div 6$. Count by sixes — 6, 12, 18, 24, 30, 36, 42 — that is seven sixes. Proof by multiplication: $6 \times 7 + 3 = 42 + 3 = 45$. The baker filled 7 bags.

4. Estimate first: $632 \div 8$. An estimate is easiest when the dividend is a number eight divides neatly, so round first. 632 is between 600 and 700; the tens and ones make 32, which is less than half of a hundred, so 632 rounds down to 600. Now divide the friendly number: $600 \div 8$. Since $8 \times 75 = 600$, the estimated quotient is 75. Each member should get about that many tickets — the exact answer is 79, so the estimate is in the right neighbourhood.

5. Vans for the field trip: 53 students, 8 per van. Divide: $8 \times 6 = 48$ students seated, and $53 - 48 = 5$ students still standing. Since 5 is smaller than 8, six vans is as far as whole vans go: quotient $\boxed{6}$ full vans, remainder $\boxed{5}$ students.

Now read the question again. The five leftover students still have to travel, and part of a van cannot be booked, so the remainder forces one more whole van. The organiser must book $\boxed{7}$ vans.

6. Share the marbles: 74 marbles among 9 friends. Dealing them out one at a time, each round of dealing uses nine marbles. Eight rounds use $9 \times 8 = 72$ marbles, and $74 - 72 = 2$ marbles are left — too few to give everybody another one, since two friends would get a marble and seven would not. Check: $9 \times 8 + 2 = 74$. Each friend gets $\boxed{8}$ marbles and $\boxed{2}$ are left over.

7. Partial quotients: 145 stickers, 6 per page. Take away easy chunks of pages instead of one page at a time. Twenty pages hold $6 \times 20 = 120$ stickers, leaving $145 - 120 = 25$. Four more pages hold $6 \times 4 = 24$, leaving $25 - 24 = 1$. Add the chunks of pages: $20 + 4$ pages, and one sticker will not fill a page, so it is the remainder. Check: $6 \times 24 + 1 = 144 + 1 = 145$. Quotient $\boxed{24}$ pages, remainder $\boxed{1}$ sticker.

8. Find the mistake: Nia wrote $87 \div 4 = 20 \text{ R } 7$. The remainder rule settles it without redividing: a remainder must be *smaller* than the divisor, and 7 is bigger than 4. A remainder of 7 still contains another whole group of four, so Nia stopped one group early — her answer describes only 20 of the groups she could have made. Take that extra group: one more four leaves $7 - 4 = 3$, so the quotient goes up by one and the remainder becomes three. Check by multiplying back: $4 \times 21 + 3 = 84 + 3 = 87$. The correct quotient is $\boxed{21}$ and the correct remainder is $\boxed{3}$.

9. Estimate, then divide: 1347 books, 5 per shelf. Round to the nearest hundred first. 1347 sits between 1300 and 1400, and 47 is less than half a hundred, so it rounds down to $\boxed{1300}$. Estimate: $1300 \div 5 = \boxed{260}$ shelves, because $5 \times 260 = 1300$. Now the exact division. $5 \times 269 = 1345$, and $1347 - 1345 = 2$, with 2 smaller than 5, so the division stops there. Check: $5 \times 269 + 2 = 1347$. The exact quotient is $\boxed{269}$ shelves with a remainder of $\boxed{2}$ books. The estimate of 260 was close, which says the exact work is believable.

10. Ribbon for bows: 96 centimetres of ribbon, 7 centimetres per bow. Count sevens: $7 \times 13 = 91$ centimetres used, and $96 - 91 = 5$ centimetres remain. Since 5 is less than 7, no further whole bow can be cut. Check: $7 \times 13 + 5 = 96$. Whole bows: $\boxed{13}$, ribbon left over: $\boxed{5 \text{ cm}}$.

The explanation (accept any answer with this idea — open response, reviewed by an adult): in the van story the leftover people still have to be carried, so the leftover forces an extra van and the answer is rounded up. In the ribbon story the leftover is only material: five centimetres is not enough for a bow, so it is simply left over and the answer is rounded down. The arithmetic $96 \div 7 = 13 \text{ R } 5$ is identical in both stories — what changes is the question, so the remainder has to be read, not just reported.